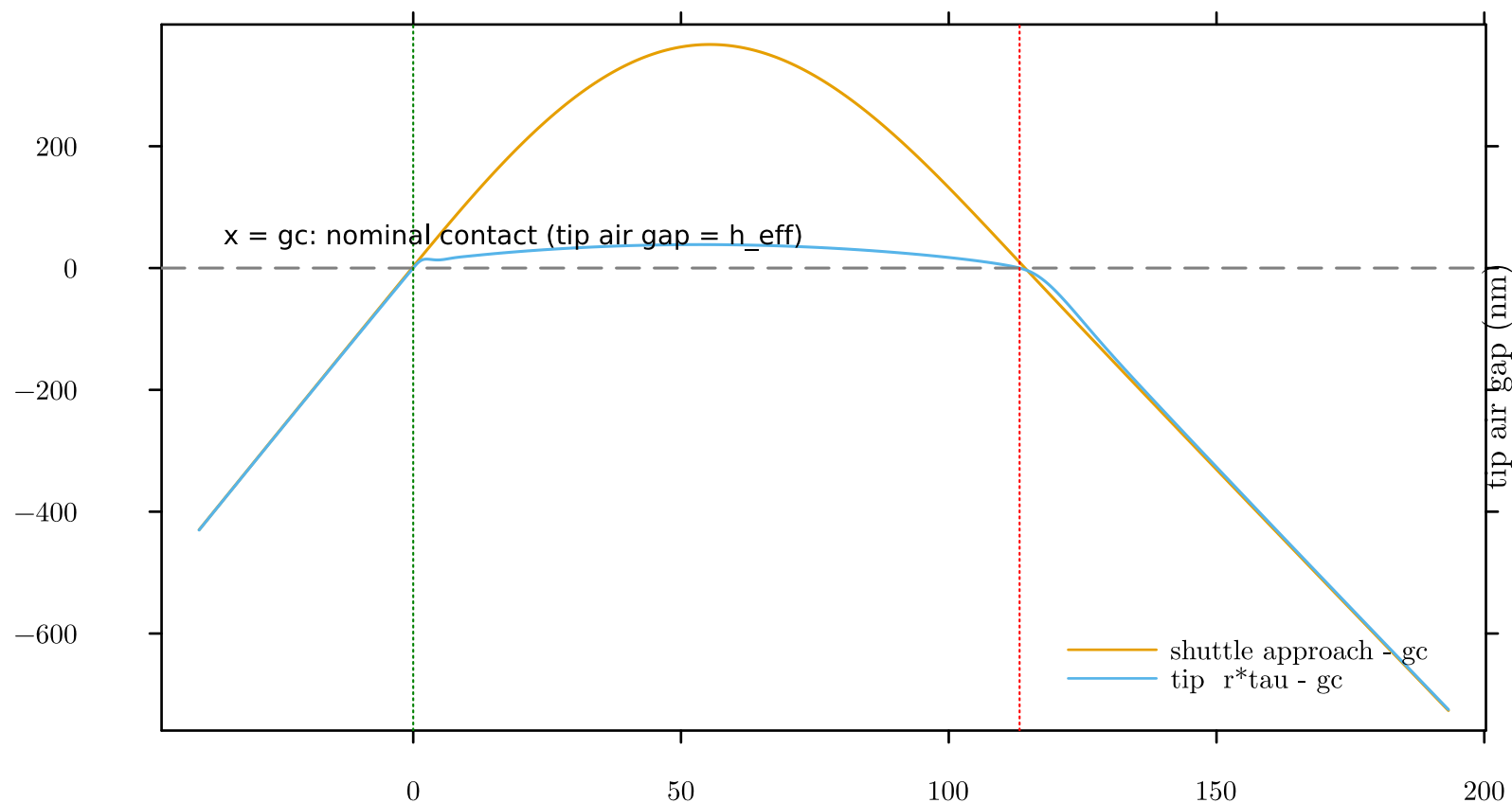
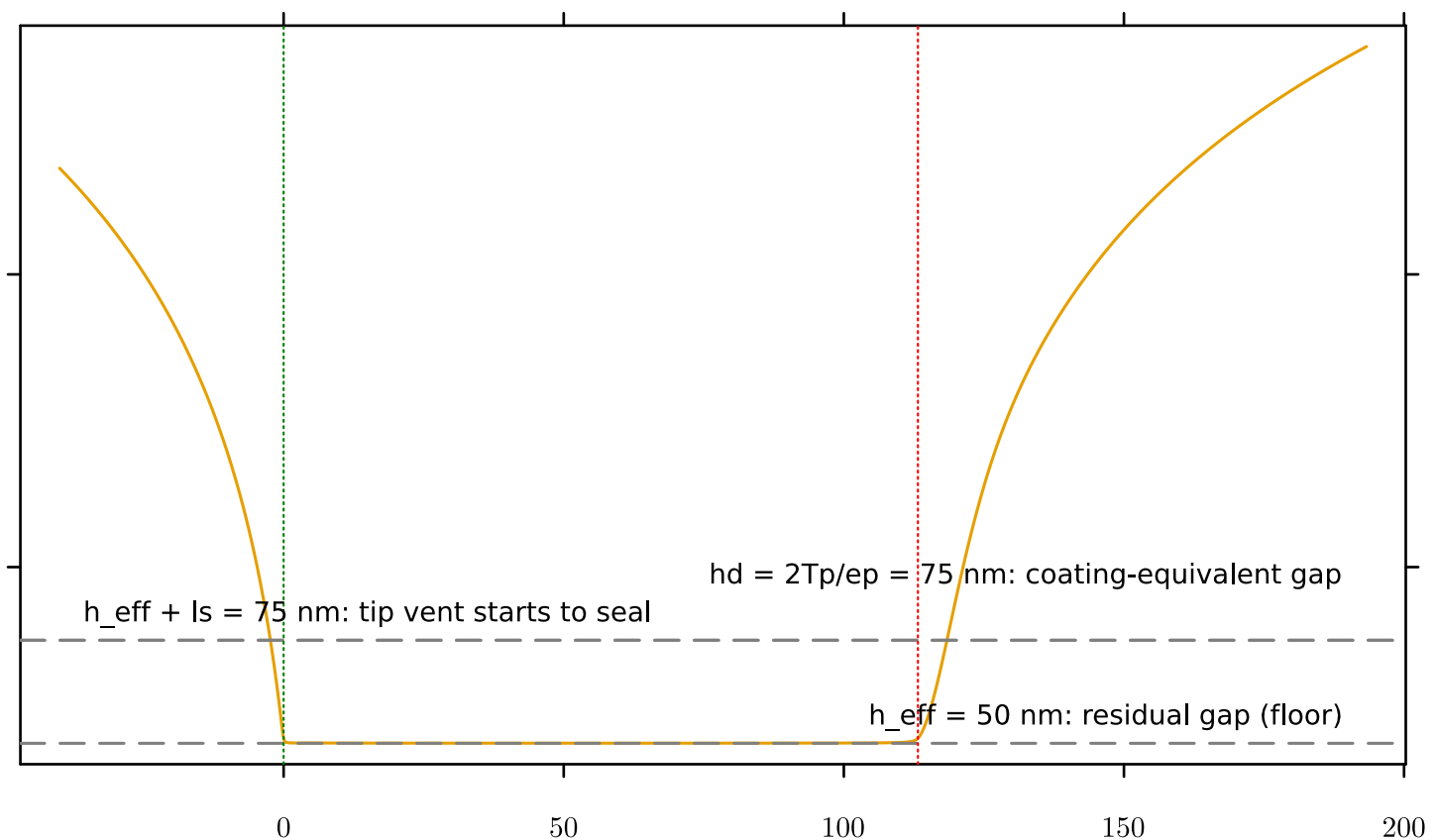


Collision close-up - hardest impact, $t = 0.04735$ s, wall -1 (green / red dotted: contact made / lost)

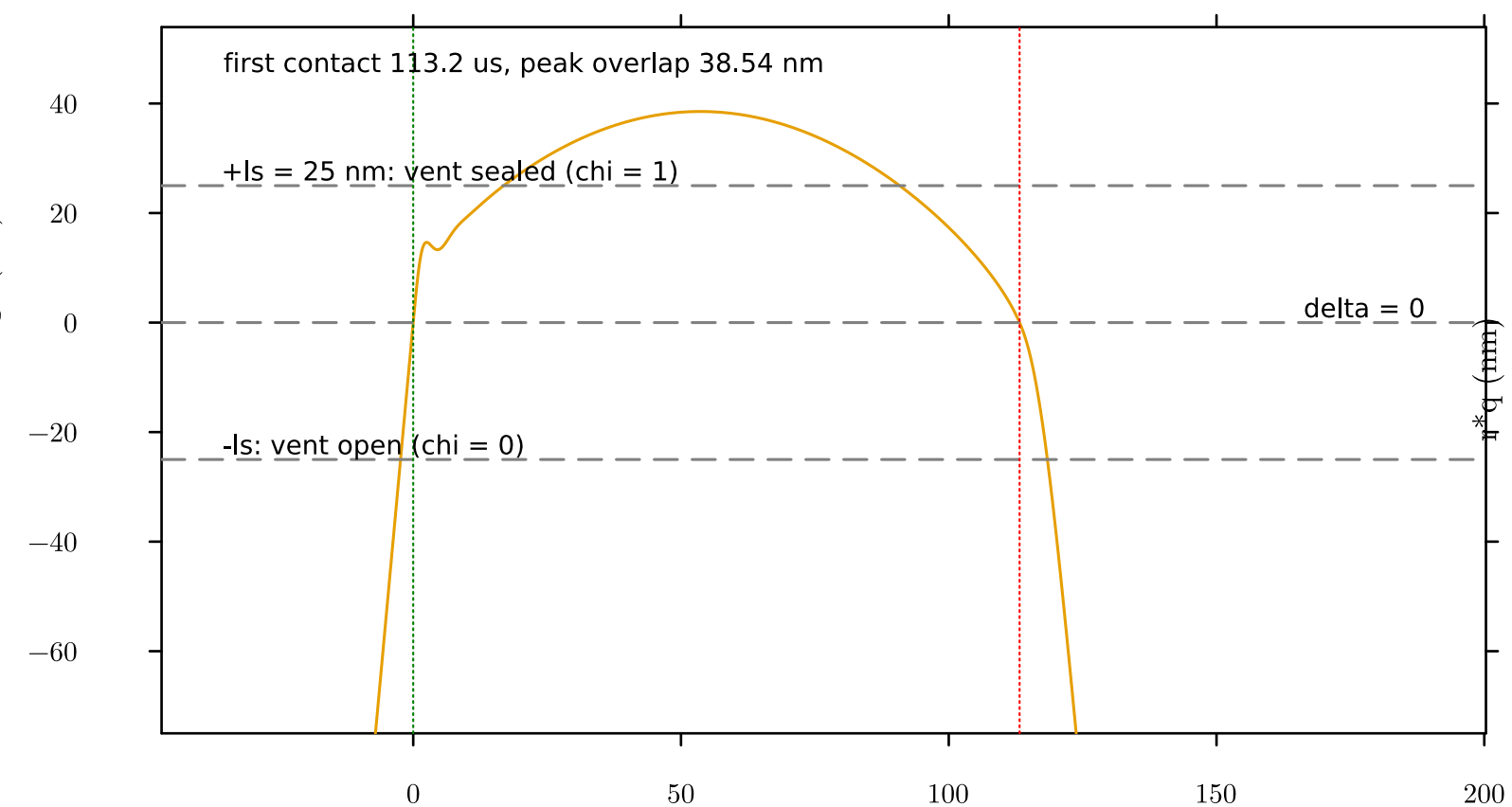
Shuttle and tip relative to the contact boundary



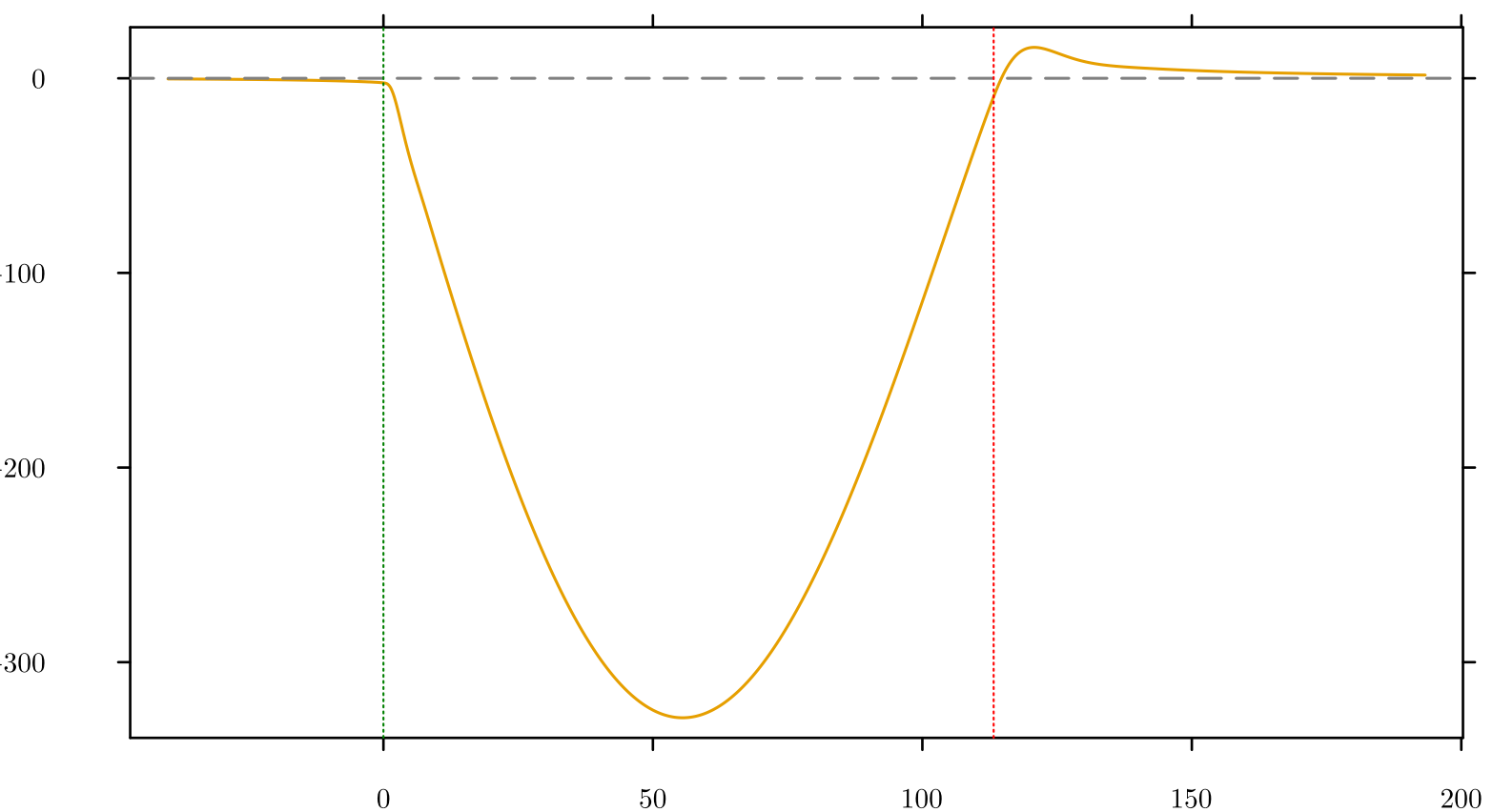
Gap closure: $h(\text{tip}) = h_{\text{eff}} + \text{softpos}(gc - r^*\tau)$



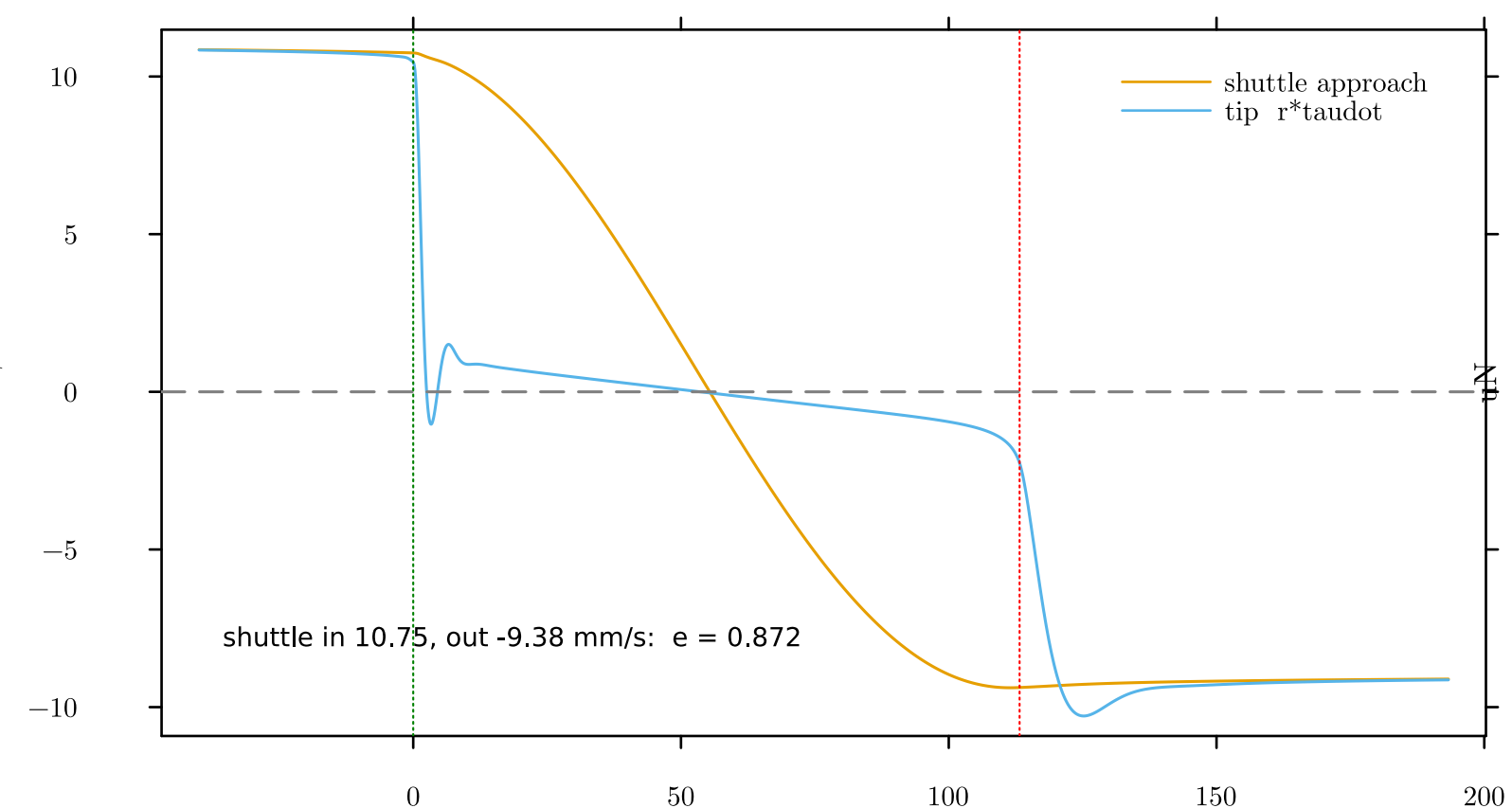
Nominal overlap and the tip-vent sealing window



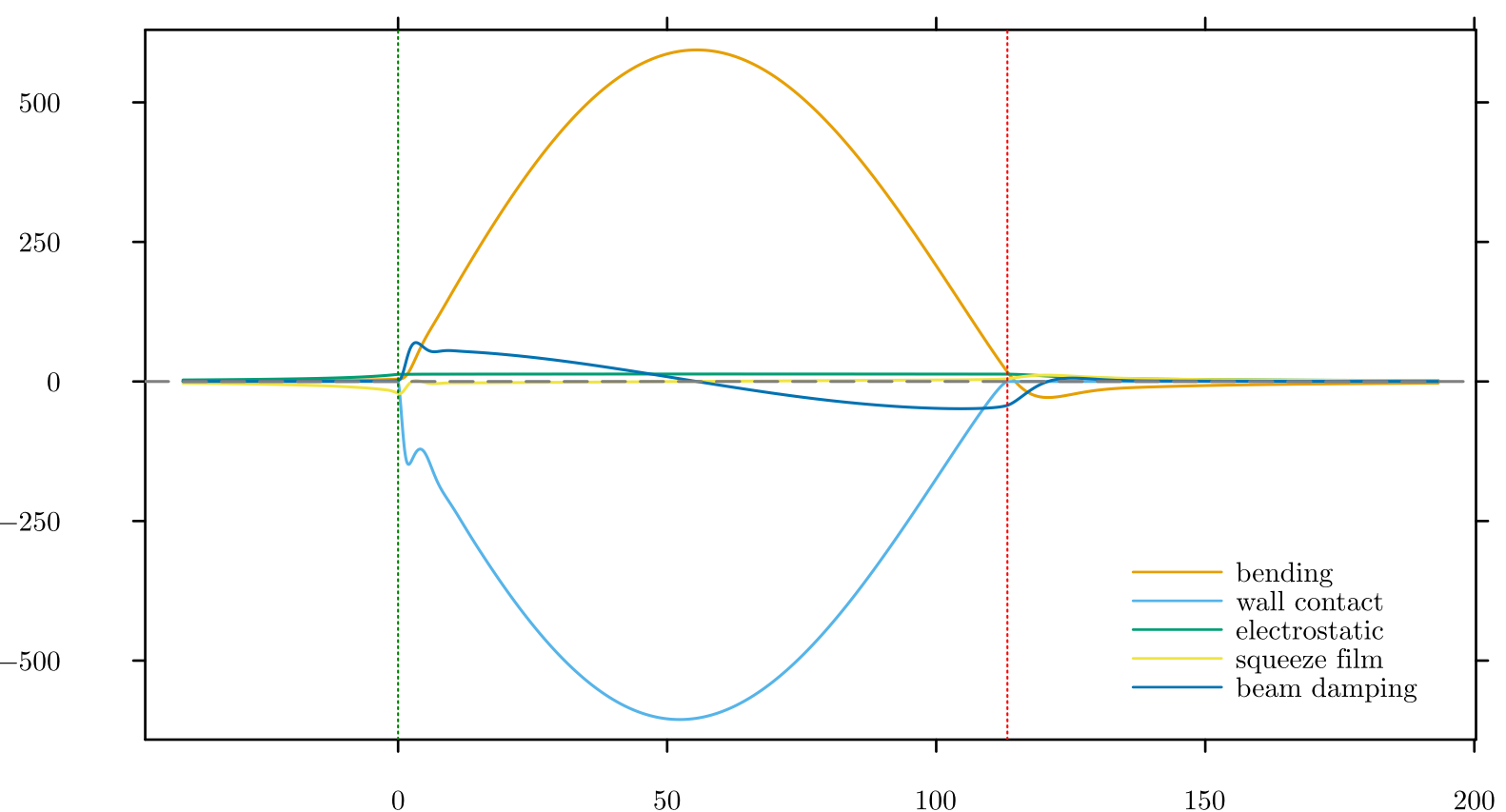
Bending of the compliant electrodes



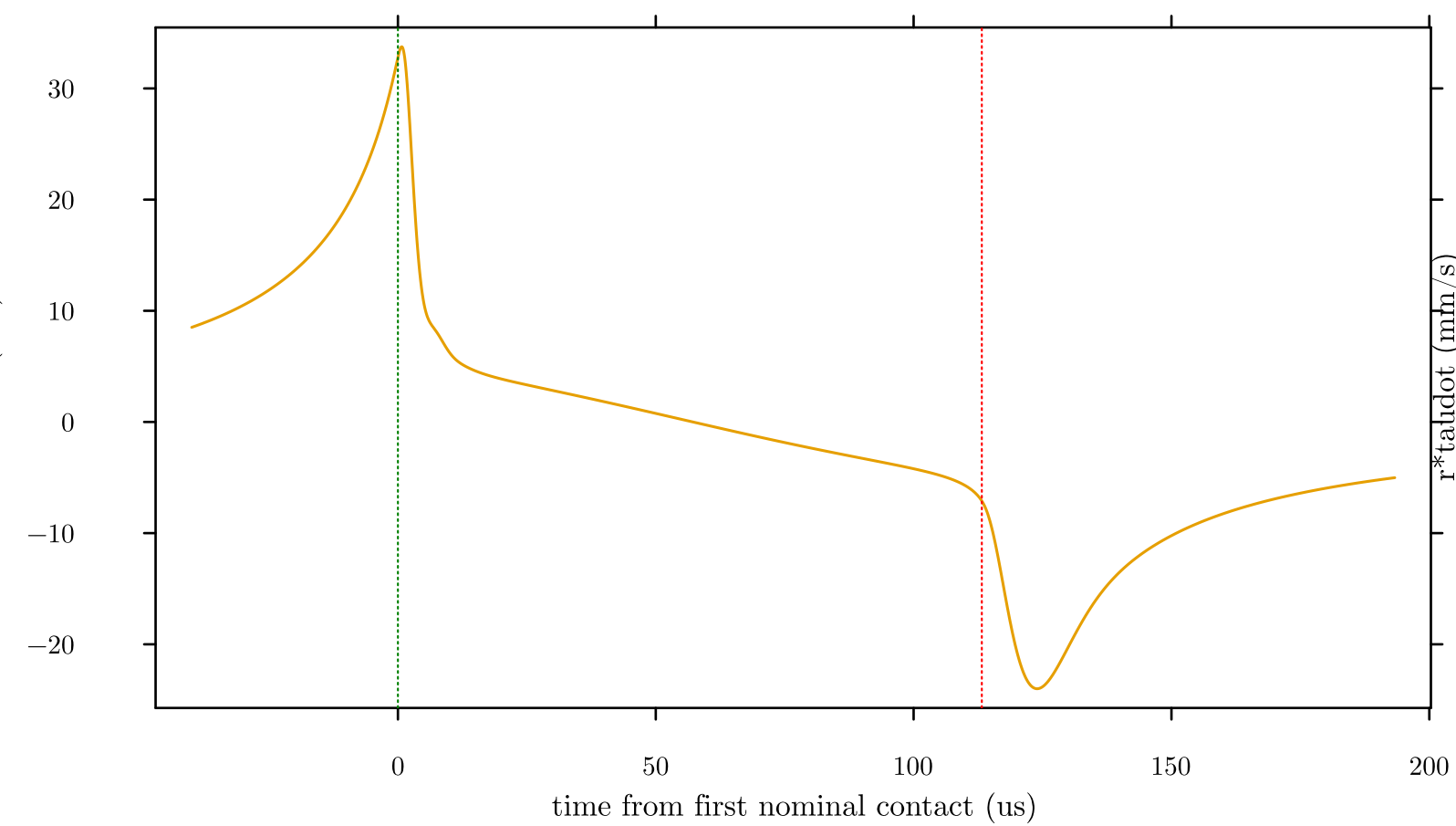
Velocities normal to the wall



Forces on the tip coordinate (+ = into the wall)



Output voltage



Phase portrait at the contact boundary

